



LECTURE 7:

ADVANCED CNN ARCHITECTURES

University of Washington, Seattle

Fall 2025



OUTLINE

Part 1: Limitations of conventional CNNs

- Very deep networks
- Degradation problem

Part 2: Additional CNN components

- Residual block
- Bottleneck block
- Inception modules

Part 3: Final project guides

- Relevance
- Data Availability
- Domain Knowledge
- Usefulness
- Feasibility
- Possible limitations



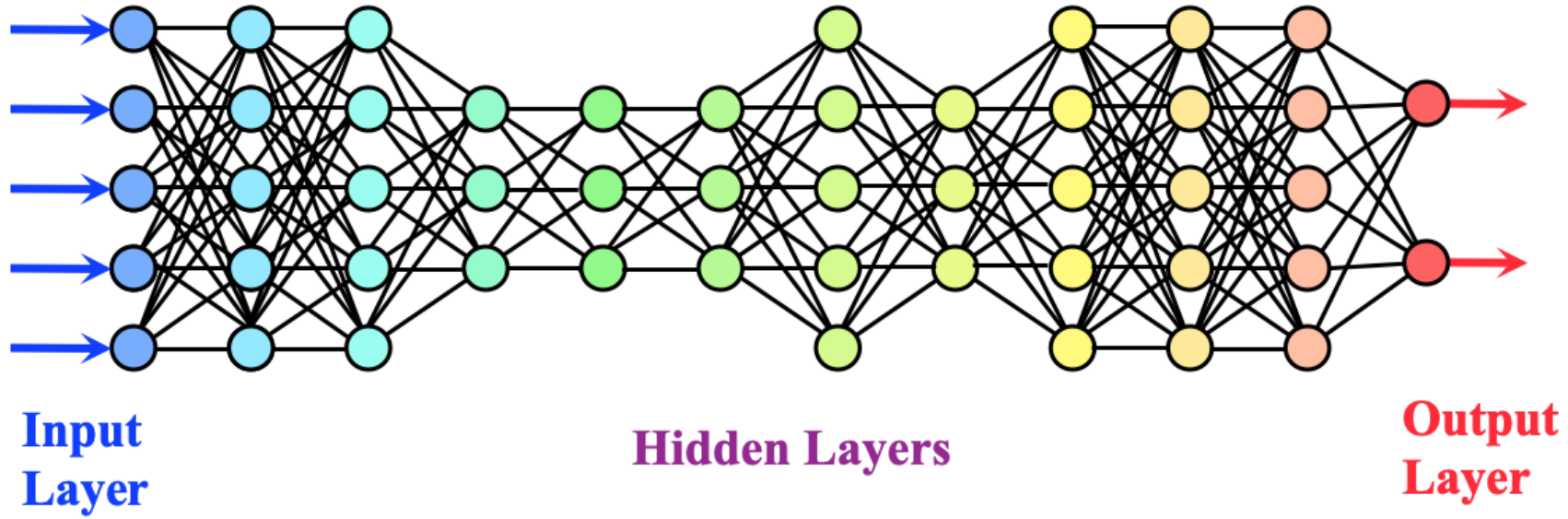
Limitation of conventional CNNs

Very deep networks

Degradation problem

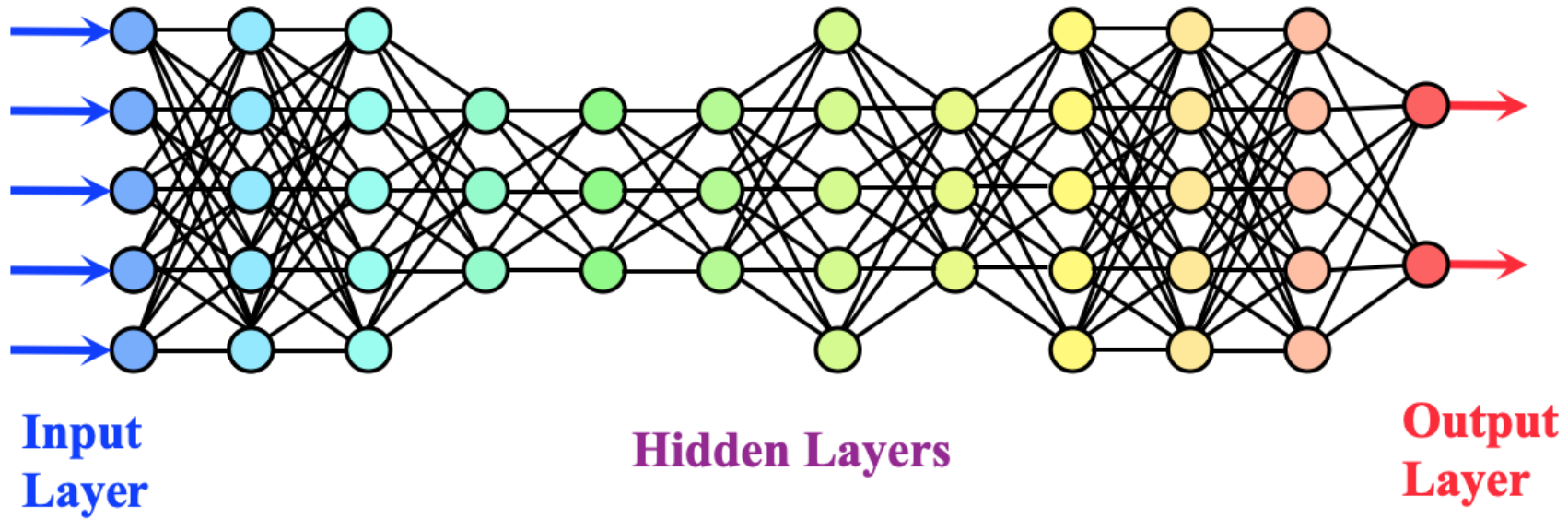


Very Deep Networks





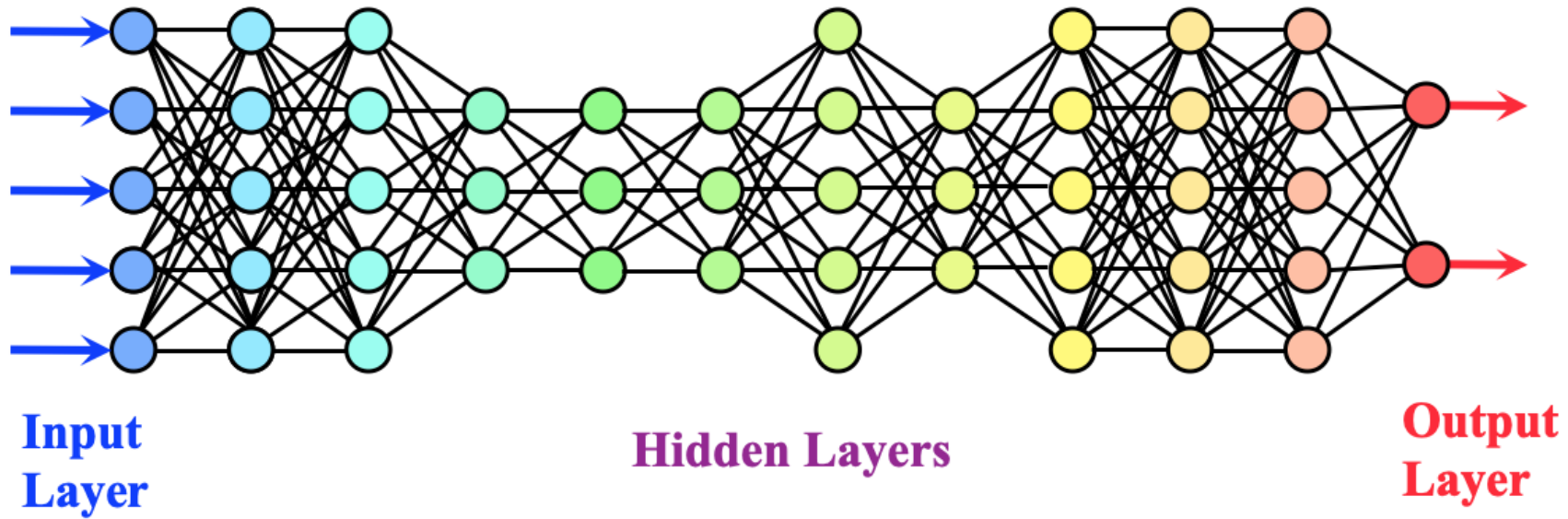
Very Deep Networks



>100 layers



Very Deep Networks



>100 layers

More layers → can process more complex patterns



Degradation problem

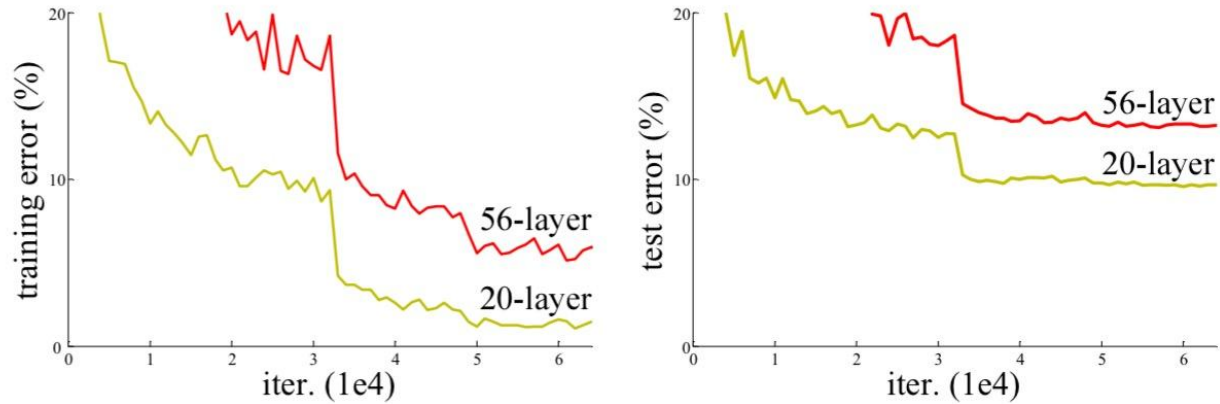


Figure 1. Training error (left) and test error (right) on CIFAR-10 with 20-layer and 56-layer “plain” networks. The deeper network has higher training error, and thus test error. Similar phenomena on ImageNet is presented in Fig. 4.

A Review of the Evolution of Deep Learning Architectures and Comparison of their Performances for Histopathologic Cancer Detection



Degradation problem

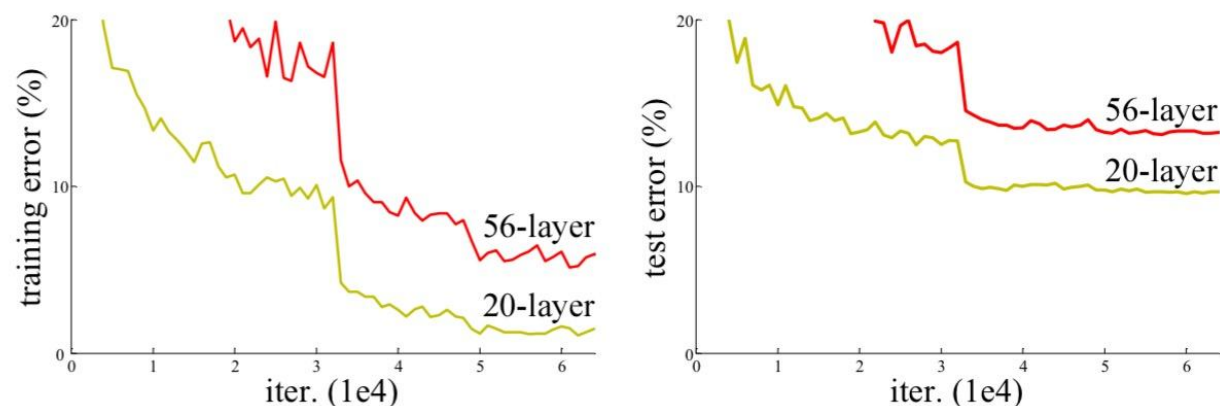
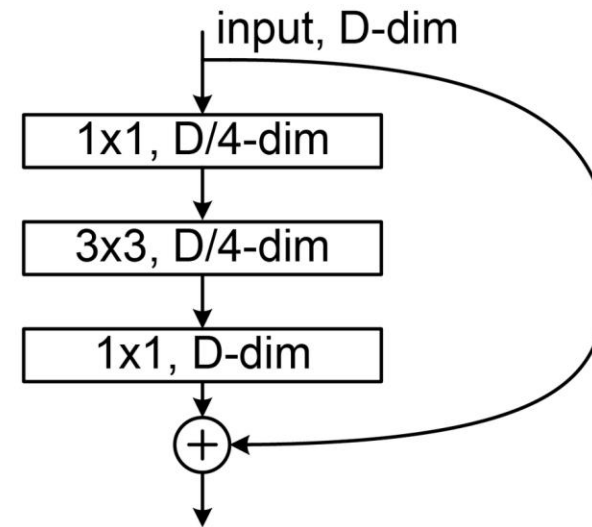
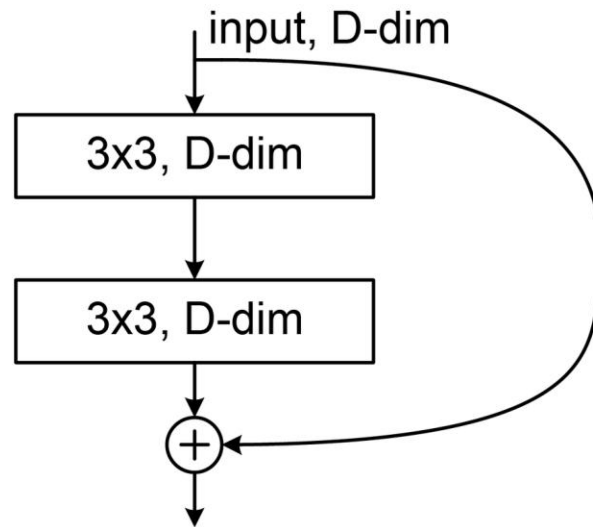


Figure 1. Training error (left) and test error (right) on CIFAR-10 with 20-layer and 56-layer “plain” networks. The deeper network has higher training error, and thus test error. Similar phenomena on ImageNet is presented in Fig. 4.

Training / Testing accuracy start saturating faster than shallow networks

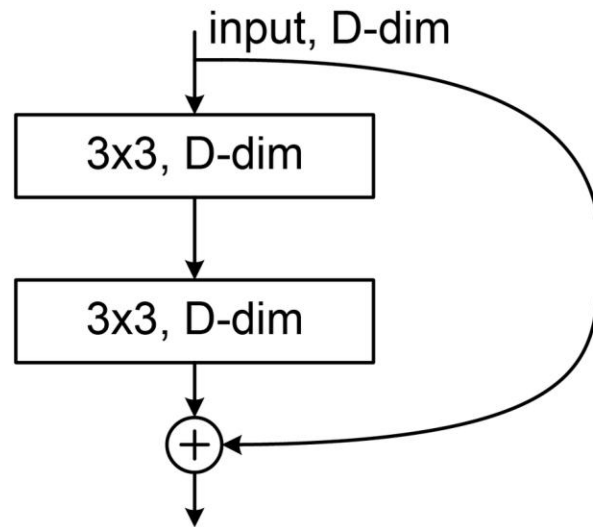


How to train very deep networks

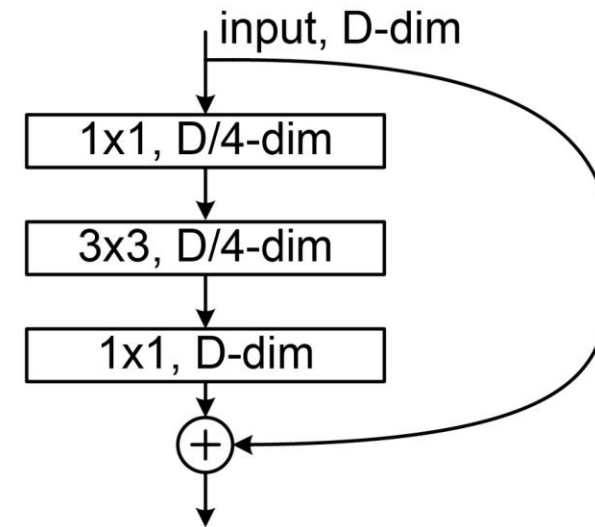




How to train very deep networks



Skip connections
(Residual block)



Data compression
(Bottleneck layer)



Additional CNN Blocks

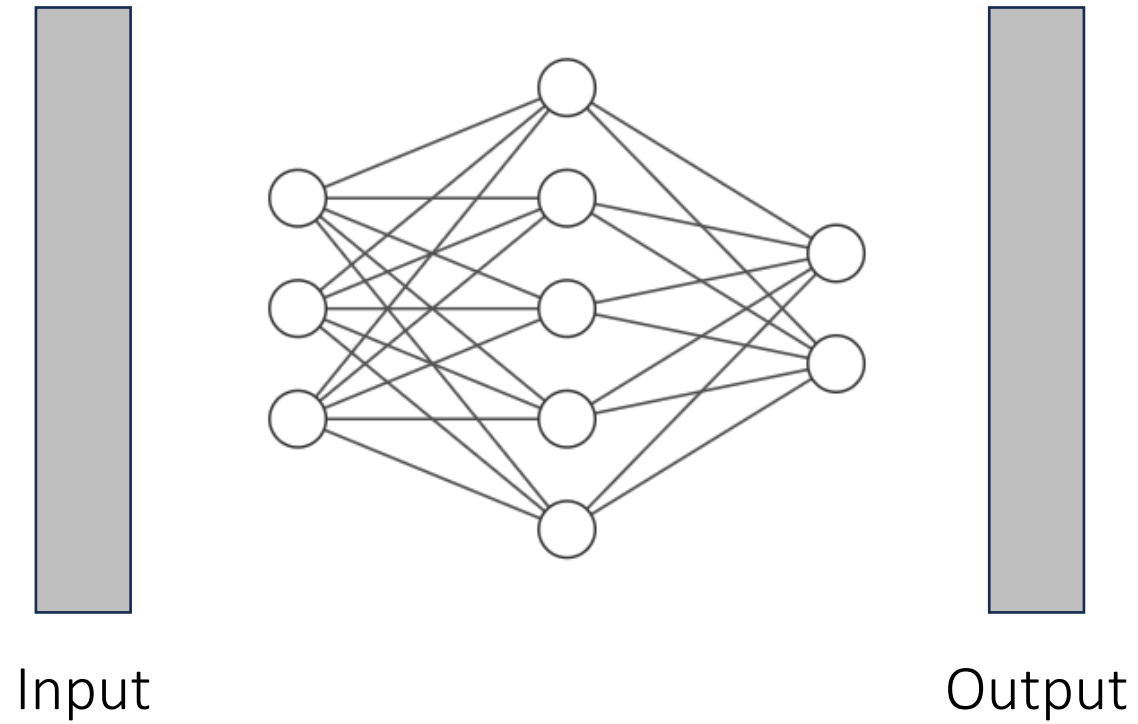
Residual blocks

Bottleneck blocks

Inception modules

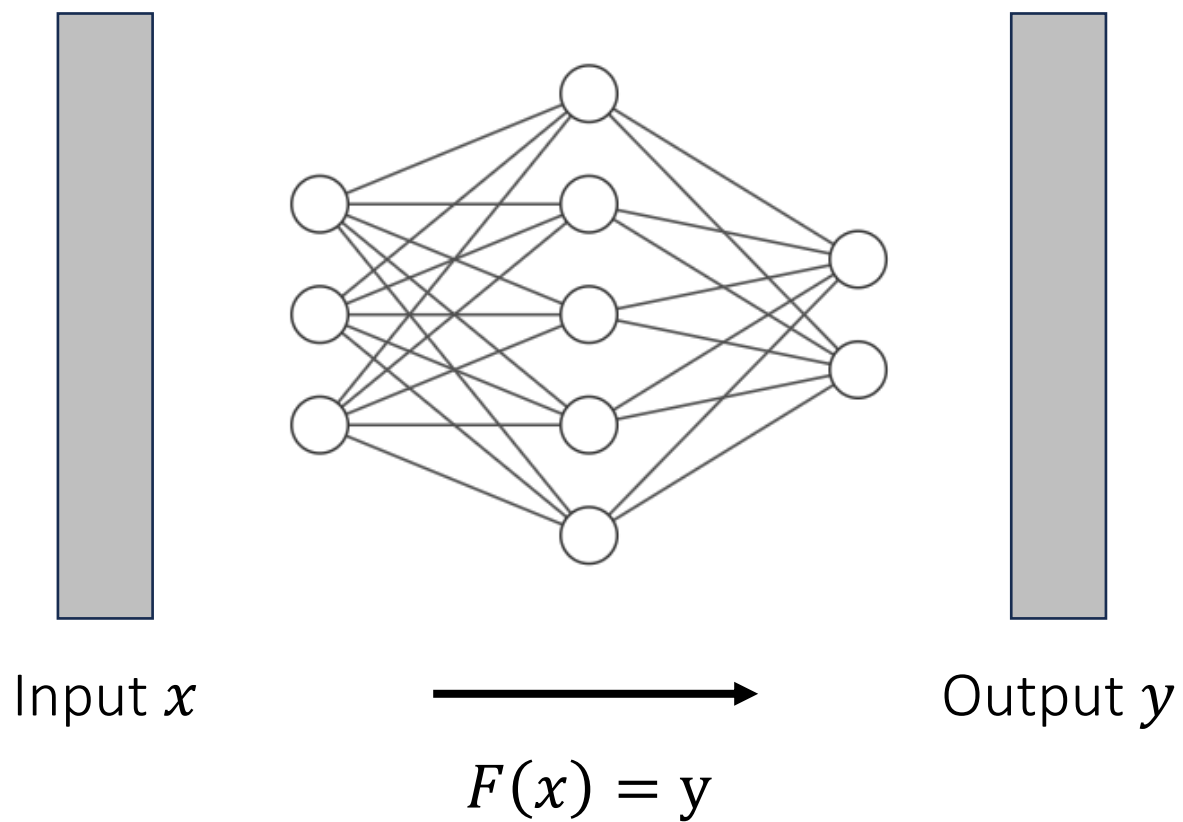


Residual blocks



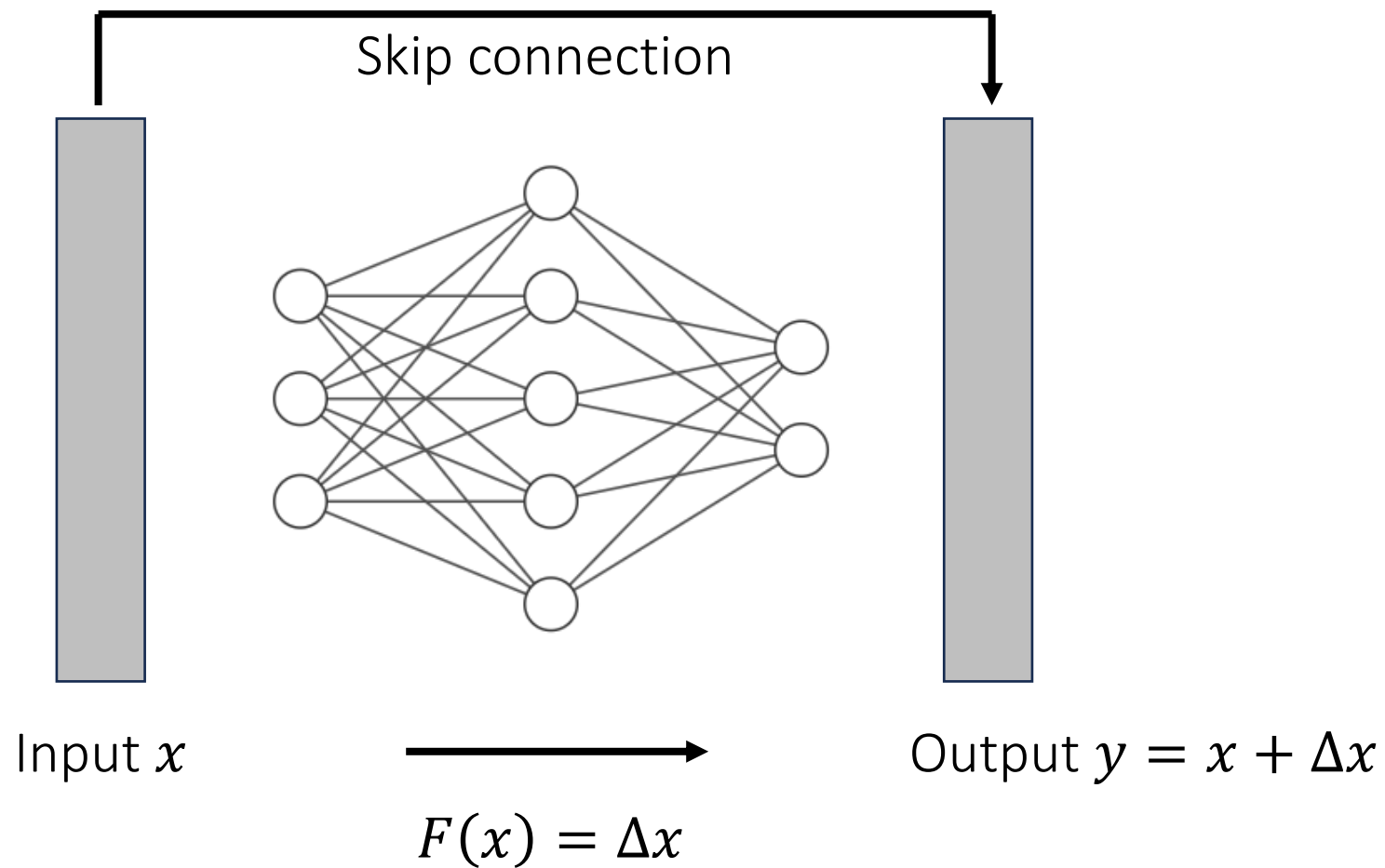


Residual blocks



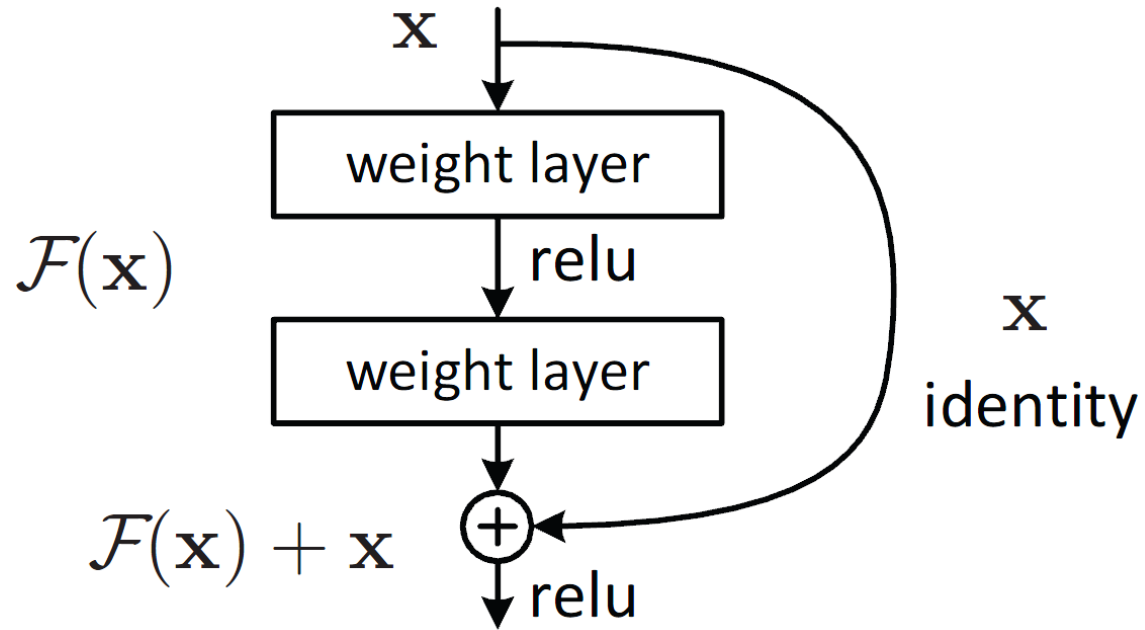


Residual blocks





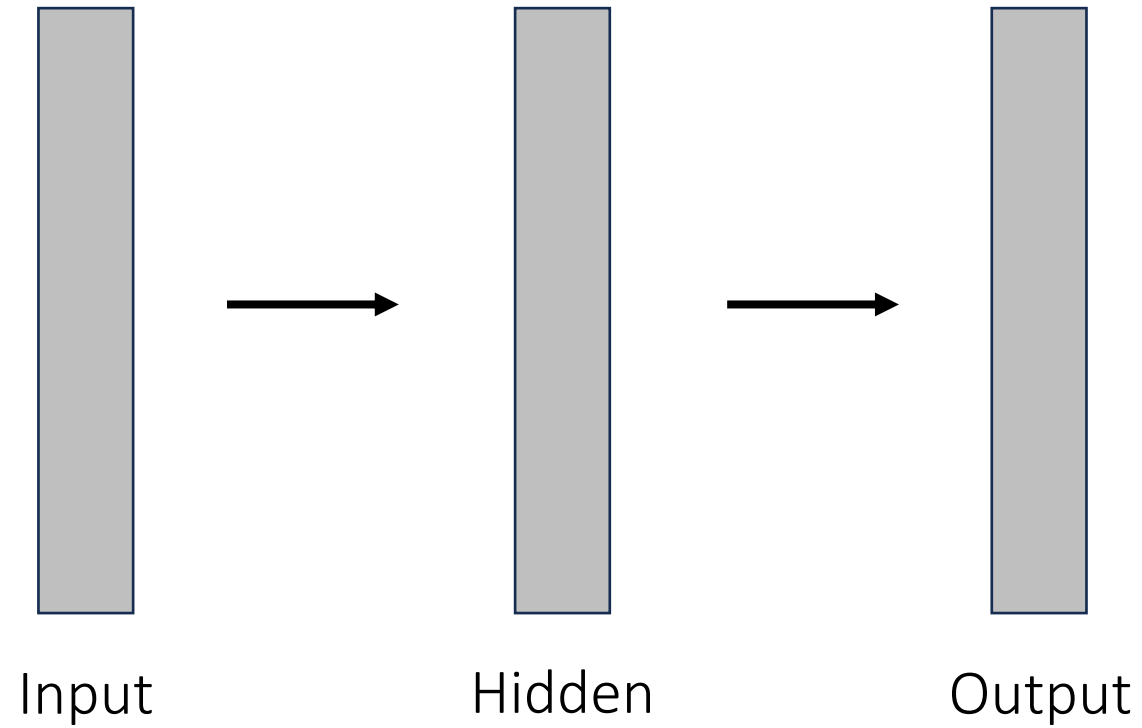
Residual blocks



- Enable training deeper networks
- Prevents vanishing gradients
- Reduces overfitting

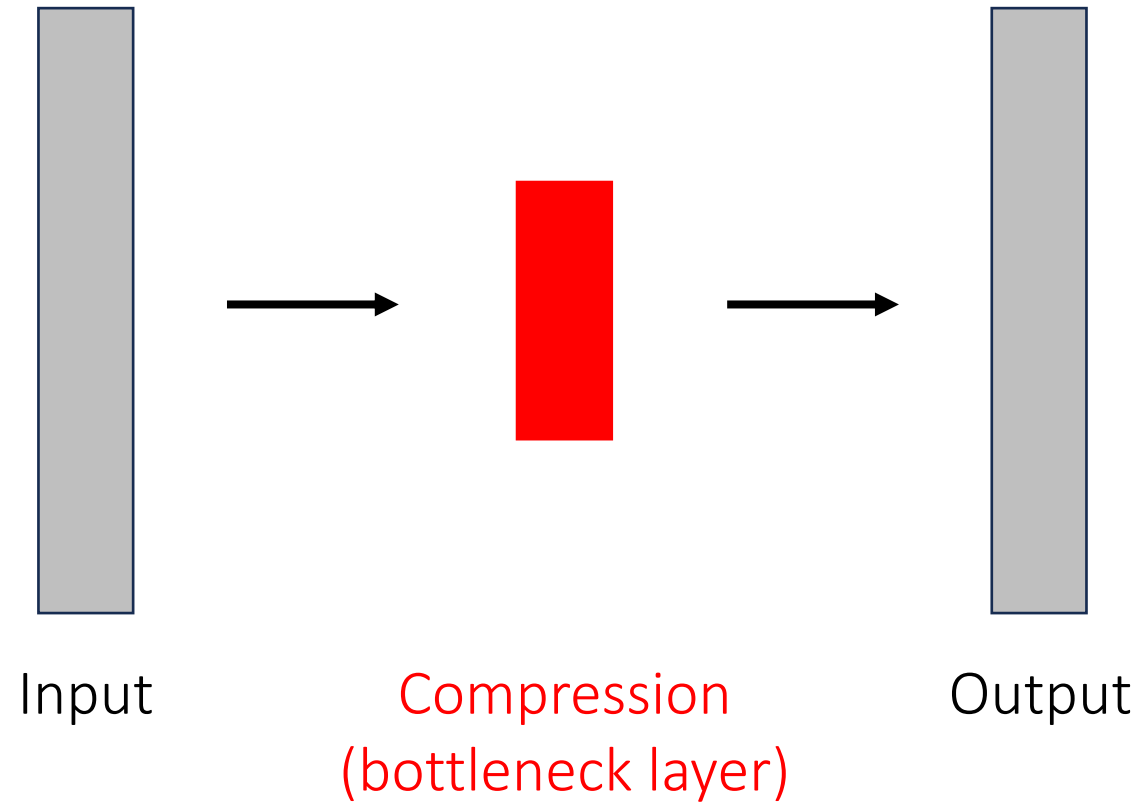


Bottleneck layer



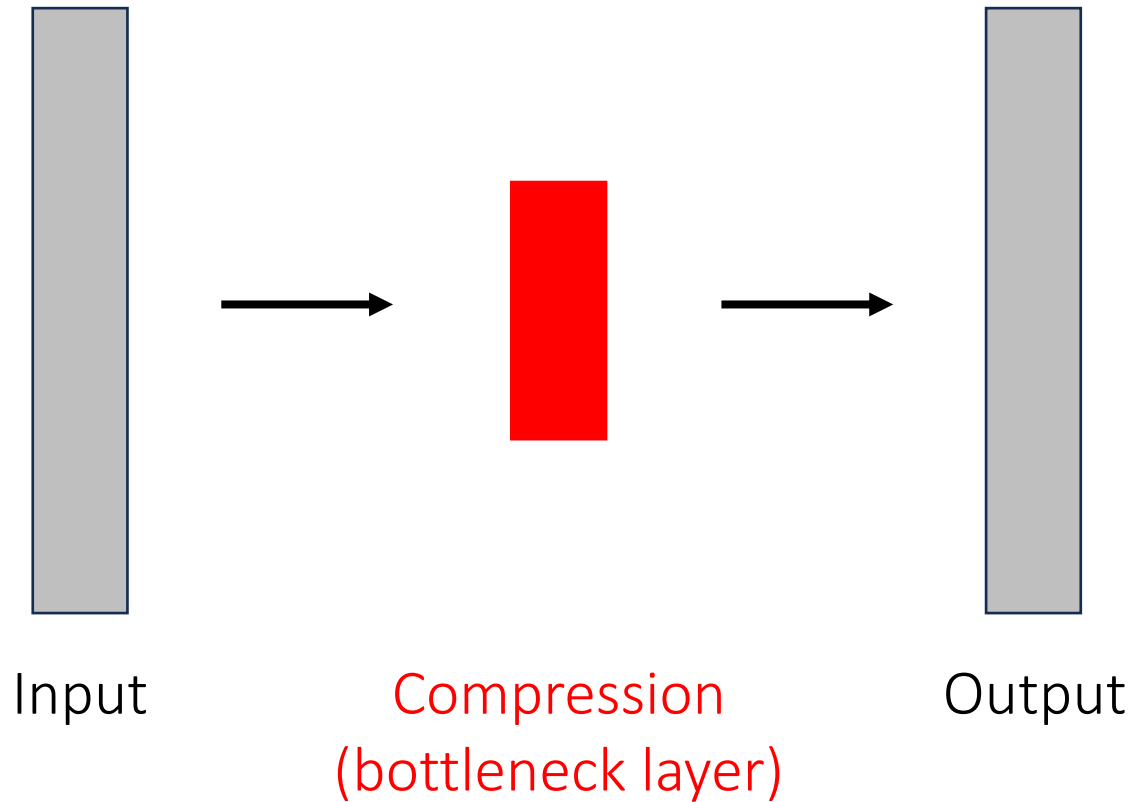


Bottleneck layer





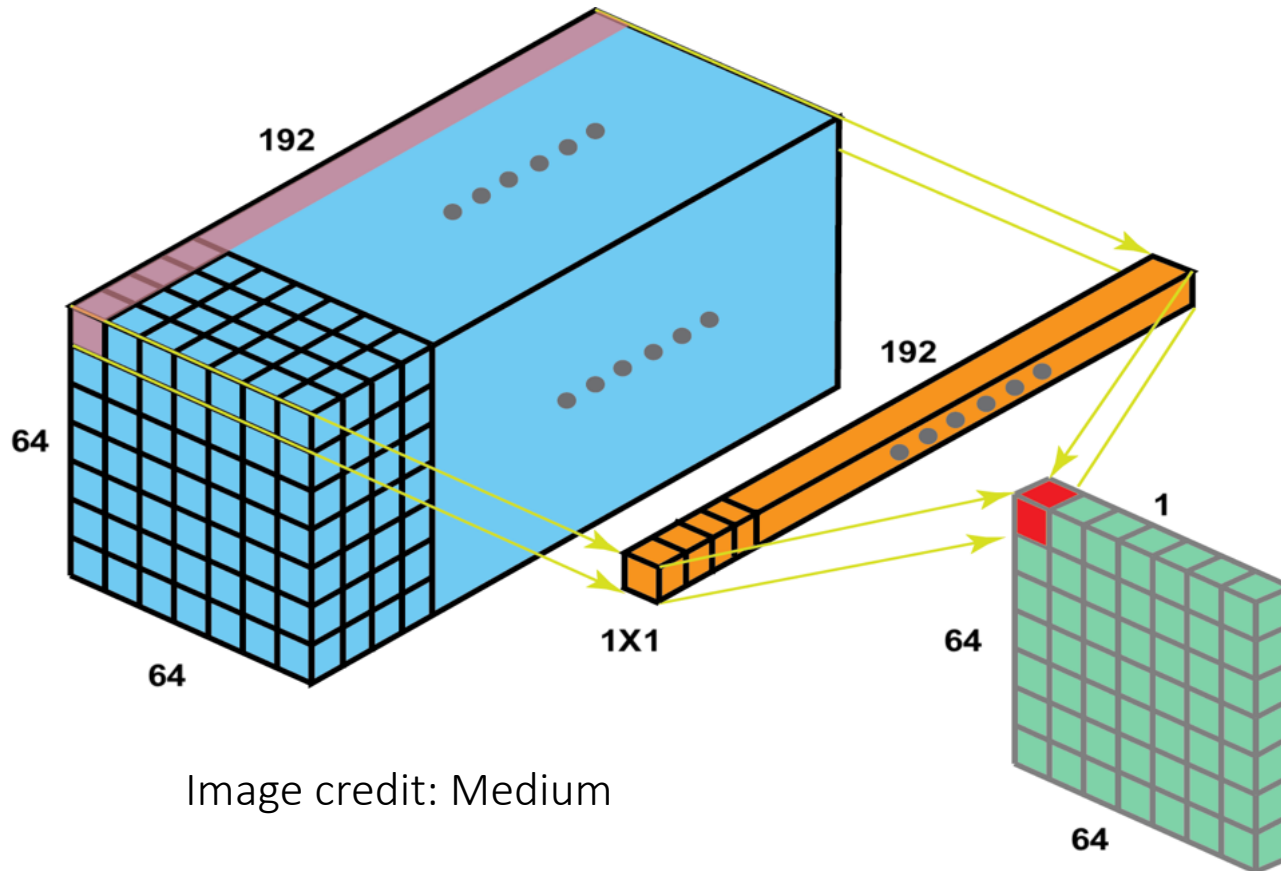
Bottleneck layer



- Low dimensional representation
- More computationally efficient (Less parameters)
- Reduces overfitting



Bottleneck layer in CNNs



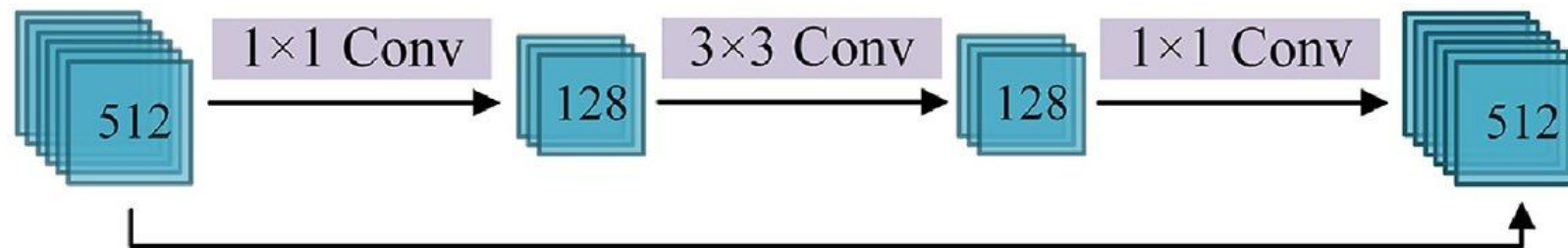
- 1 x 1 convolution allows compression in channel direction
- Similar to pooling but adjusting dimension in channels
- Acts as linear combination of the input channels



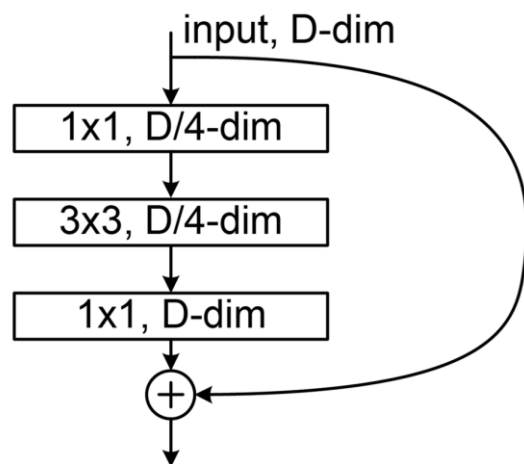
Bottleneck layer in CNNs

ResNet

Zhao et al., 2019



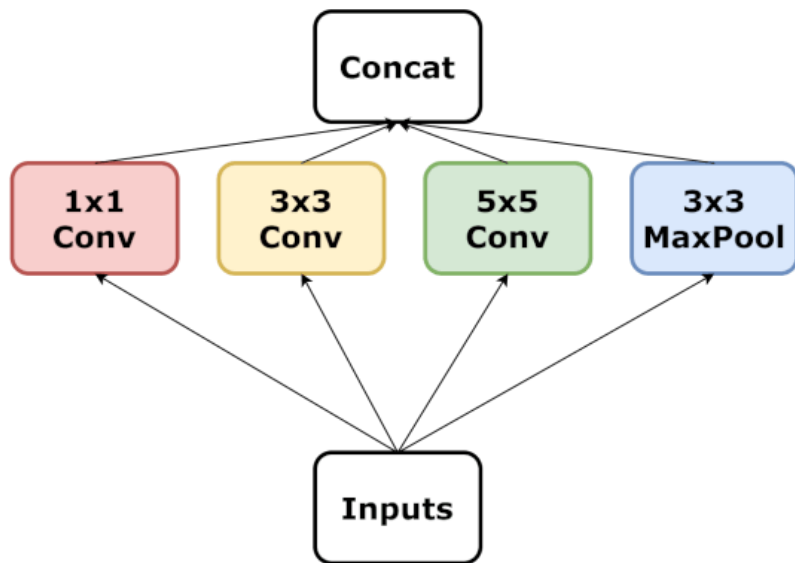
sum
512



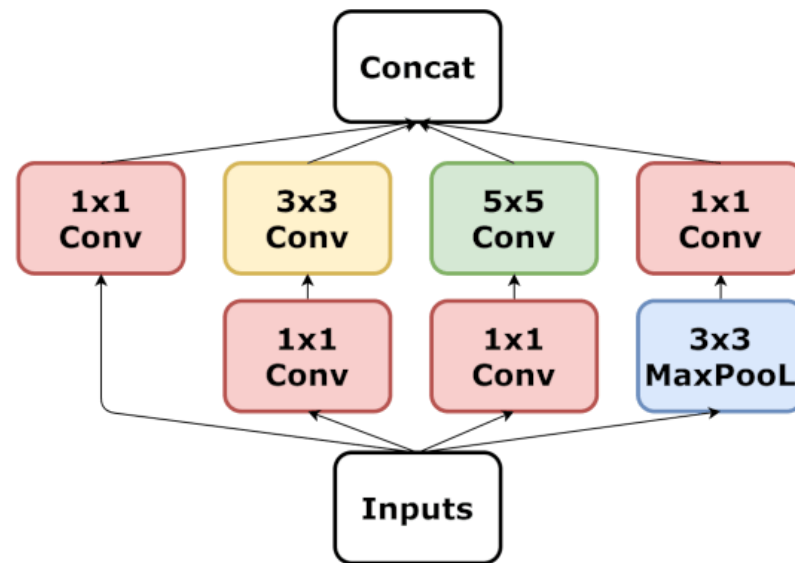
Compression
(bottleneck layer)



Inception module



Inception Module



Inception Module with Dimension Reduction

Szegedy et al., 2014



Inception module

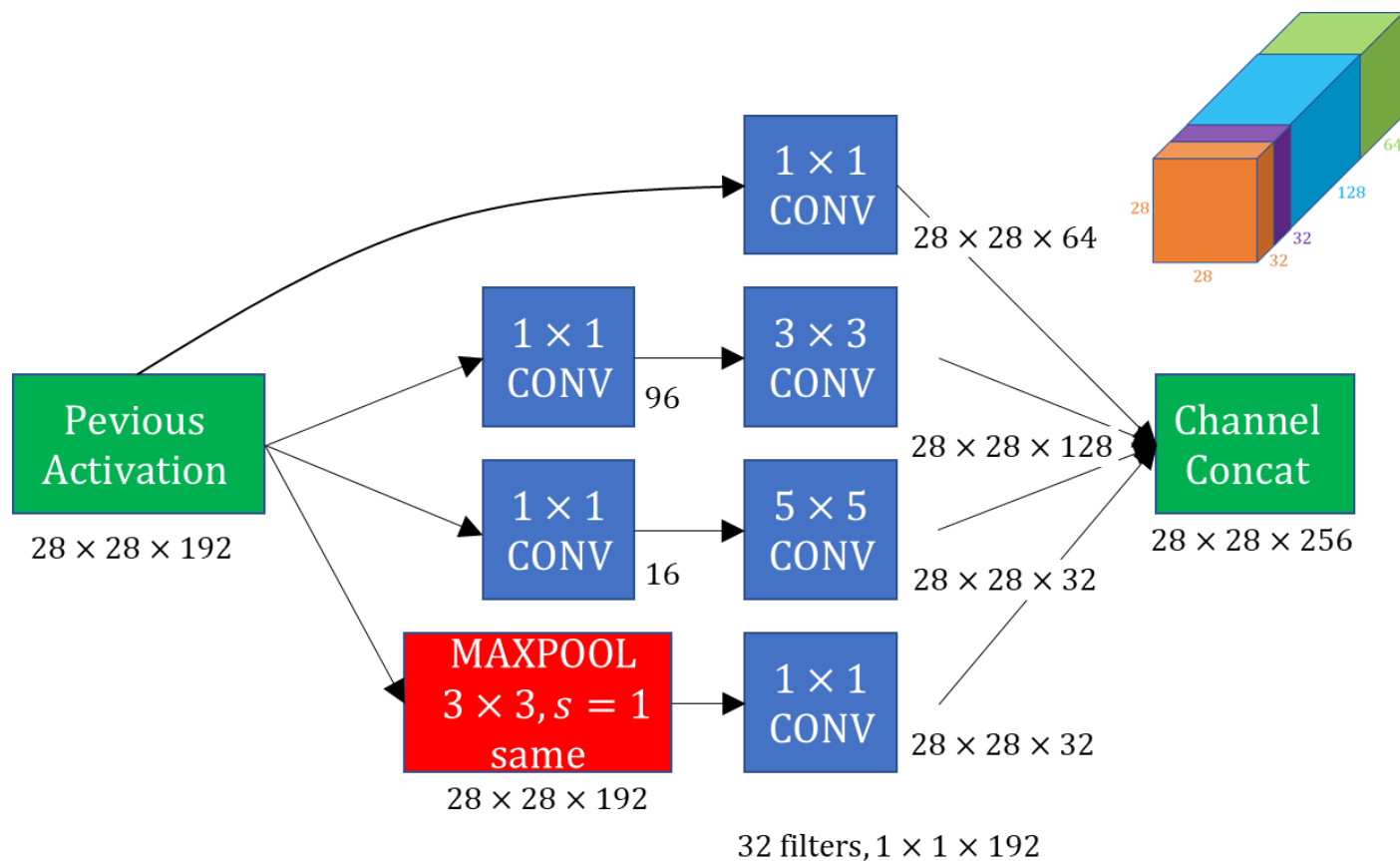
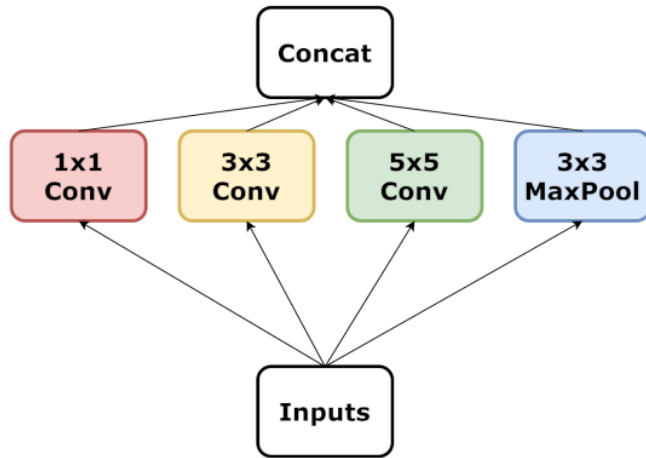


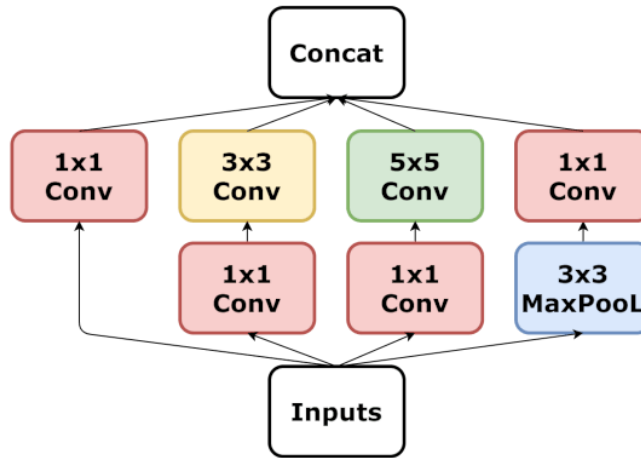
Image credit: Master Data Science



Inception module



Inception Module

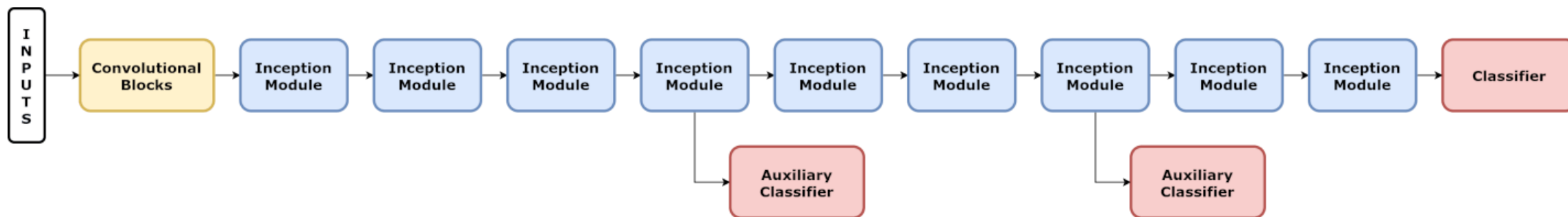


Inception Module with Dimension Reduction

- Parallel convolution operations
- Richer variety of features in a single output
- Can be used as standardized module



Inception module



Inception V3 model



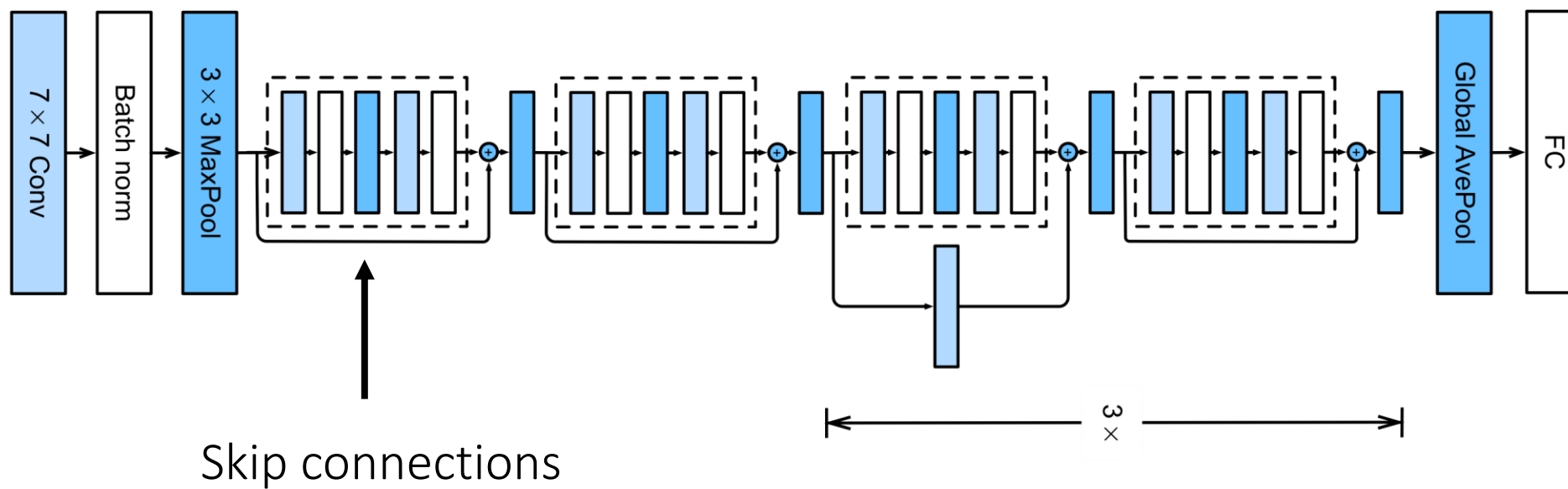
Advanced CNN examples

ResNet-18

GoogLeNet

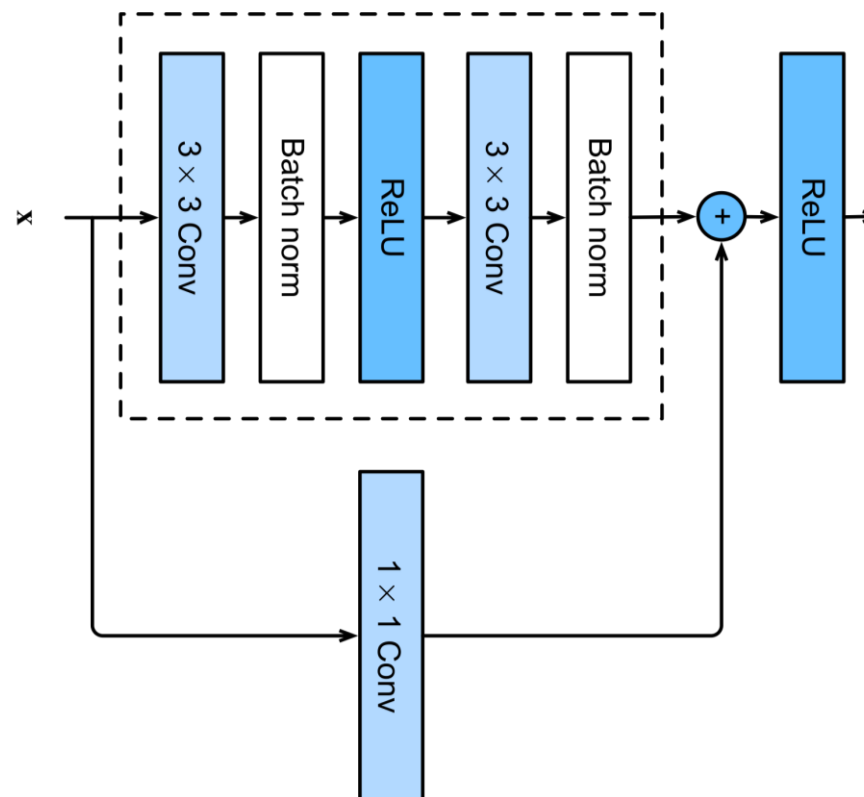
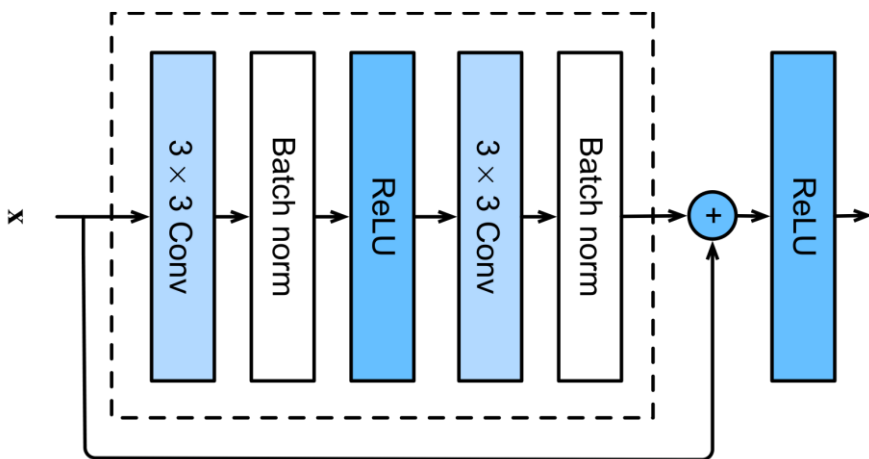


ResNet-18





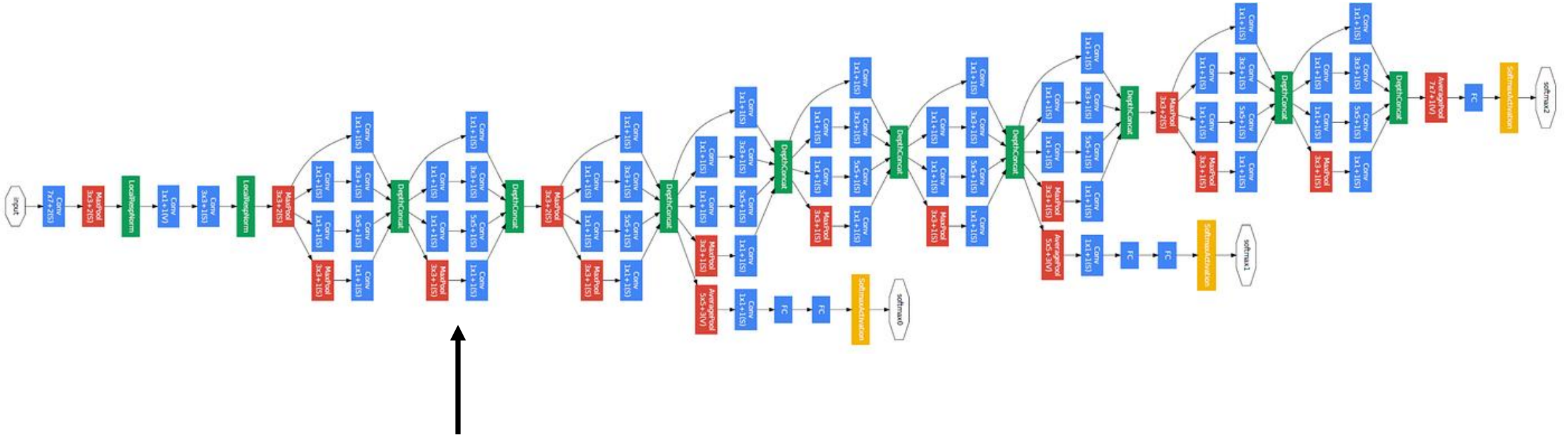
ResNet-18



1 x 1 conv for down-sampling



GoogLeNet



Inception module



Final Project Guide



Final Project Checklist

Picking the problem

Relevance

Data Availability

Domain Knowledge

Usefulness

Feasibility

Possible Limitations and Concerns



Picking the Problem

Method type

Supervised

Unsupervised

Reinforcement

Input

Data with
labels

Data without
labels

States + Actions

Output

Mapping

Classes

Action



Relevance and Interest

Relevance

- Q: Is the project relevant to the course (i.e., ML and DL)?

Interest

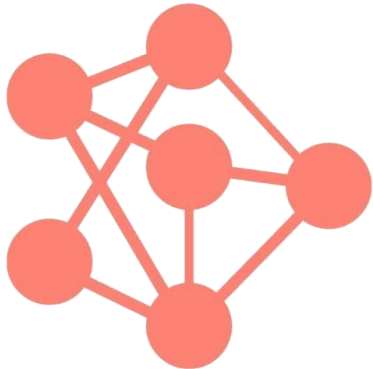
- Q: Is the project about something you are excited about?



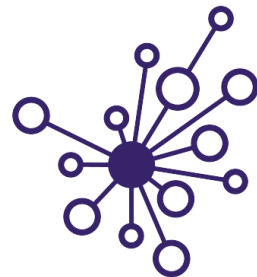
Data Availability

kaggle

NeurIPS 2024 Competition Track Program



fellowship.ai



eScience Institute



Domain Knowledge

Q: What is your (group) area of expertise?

e.g., NLP, Image processing, Signal processing, Generative tasks, Decision agent, Theoretical studies etc



Usefulness and Utility

Q: What's the significance of your project?

Q: What will be the evaluation metric?

Q: What would constitute a good performance or outcome?

Q: What are the existing methods that you can use as baseline?



Feasibility

- Balance between over-ambitious and under-ambitious
- Time management for Planning -> Execution -> Iteration
- Leave time for iterations/improvements and benchmarks



Possible Limitations and Concerns

- Failure is an option!
- What can be done to further improve the method or address issue?
- What are the method's limitations?
- Any potential ethics problems?

